

THE CLOSED CAUSAL LOOP & EPISTEMIC BOUNDARY SYNTHESIS

Unifying All 17 Chapters Across 4 Parts: Sensory Transduction, Cognitive Deliberation, Real-Time Invariant Enforcement, and Physical Dynamics

PART I: FOUNDATIONS (Ch 1–4)

Body, Brain & Nervous System

Ch 1: The Causal Boundary
Ch 2: Physical Body & Dynamics
Ch 3: Learned Brain & Latency
Ch 4: Nervous System & Clocks

PART II: DATA & LEARNING (Ch 5–7)

Demonstrations, Training & Bounds

Ch 5: Sensor Demonstration Data
Ch 6: Imitation & RL Training
Ch 7: Evaluation & Rare Events

PART III: ONLINE RUNTIME (Ch 8–13)

Perception to Edge Placement

Ch 8: Perception · Ch 9: Memory
Ch 10: Intent · Ch 11: Planning
Ch 12: Safety · Ch 13: Placement

PART IV: GOVERNANCE (Ch 14–17)

Verification, Release & Frontier

Ch 14: Shared Intervention
Ch 15: Empirical Stress Testing
Ch 16: Formal Safety Release
Ch 17: The Epistemic Frontier

THE BRAIN: COGNITIVE DELIBERATION

High-Capacity Stochastic Policy (MPU)

Cadence: 20–50 Hz · Linux / NPU · Latency τ_{infer}

- VLM Semantic Goal Grounding (Ch 10)
- SE(3) Coordinate Frame Tree & World Model (Ch 9)
- Diffusion / ACT Action Chunk Generation $H=16$ (Ch 11)
- Expiring Intent Lease Issuance t_{expire} (Ch 10)
- Epistemic Uncertainty & Silent Distribution Shift (Ch 7, 17)

EMITS: Candidate Action Trajectory Chunk $\hat{u}_{t:t+H}$

Sensor Stream y_t (DMA)

THE PROPOSAL-PERMISSION CAUSAL INTERFACE

Hardware Architectural Separation (Ch 13, 17)

- ▶ **Shared Memory TCM:**
Copy lock-free seqlock ring buffers
- ▶ **Dead-Man Leases:**
Lease t_{expire} : halts if MPU stalls > 50 ms
- ▶ **Invariant Projection:**
 $u_{t^*} = \text{argmin} \|u - \hat{u}\|^2 \text{ s.t. } L_{\text{f}} h + L_{\text{g}} h u + \alpha(h) \geq 0$
- ▶ **Authority Priority:**
Safety Reflex > Human Takeover > Learned Policy

Proposals ($\hat{u}$)

Permission Gate

THE NERVOUS SYSTEM: REFLEX ENFORCER

Deterministic Invariant Filter (MCU)

Cadence: 1000 Hz · FreeRTOS / Cortex-M · Hard Real-Time

- Real-Time Active-Set QP Solver: $h(x) \geq 0$ (Ch 12)
- Dynamic Stopping Clearance: $d_{\text{stop}}(v) \leq d_{\text{gap}}$ (Ch 2, 12)
- Lock-Free Inter-Core Mailbox & Lease Check (Ch 4, 13)
- Shared Intervention & C² Bumpless Takeover (Ch 14)
- Hardware Watchdog & ISO 13849 Safety E-Stop (Ch 15, 16)

PERMITS: Verified Real-Time Motor Commands u_{t^*}

Actuation Torques / PWM

SENSORY TRANSDUCTION & INFORMATION-AGE PIPELINE

Photons / Forces → Digital Bits (The Transduction Boundary)

- Transduction: MIPI CSI-2 CMOS, IMU gyros, optical wheel encoders (Ch 8)
- Zero-Copy Ingestion: DMA ring buffers, hardware PTP nanosecond timestamps
- Information Age Freshness: Total lag $\tau_{\text{total}} = \tau_{\text{readout}} + \tau_{\text{DMA}} + \tau_{\text{infer}} + \tau_{\text{act}}$
- Spatial Displacement Lag: Unchecked motion during delay $d_{\text{lag}} = v \cdot \tau_{\text{total}}$ (Ch 2, 17)
- Endogenous Feedback: Action u_t shifts robot pose, shaping future observations y_{t+1}

1 kHz Fast Proprioceptive Odometry

THE PHYSICAL PLANT & NEWTONIAN ENVIRONMENT ($W_t \rightarrow W_{t+1}$)

Continuous Mechanics, Conservation Laws & Irreversibility

- Inertial Momentum: Kinetic energy $E_k = 1/2 m v^2$ (No software rollback!)
- Tire/Surface Coulomb Friction Limits: $F_{\text{friction}} \leq \mu N$ (Ch 2, 17)
- Non-Smooth Contact Dynamics: Painlevé paradoxes & brittle yield (Ch 2)
- Actuator Dissipation: Thermal Joule heating $P_{\text{loss}} = I^2 R$, winding limits
- Physical State Update: $W_{t+1} = \text{Dynamics}(W_t, u_{t^*}, \text{External Disturbances})$

Physical Ground Truth W_t

CHAPTER 17 FRONTIER SYNTHESIS: THE BOUNDARY OF VERIFIABLE KNOWLEDGE

1. Detectorless Blind Spots

States S_A & S_B produce identical $y(t)$ prior to deadline t_{deadline} (e.g. internal voids, oil slicks).

2. Astronomical Exposure Limits

Proving $p \leq 10^{-9}/h$ takes $n \geq 3.0 \times 10^9 h$ (342k yrs).
Empirical sampling cannot certify open-world safety.

3. Structural Containment Mandate

Where evidence ends, nervous system clamps authority (v_{clamp} , torque limits) to render unknown harmless.